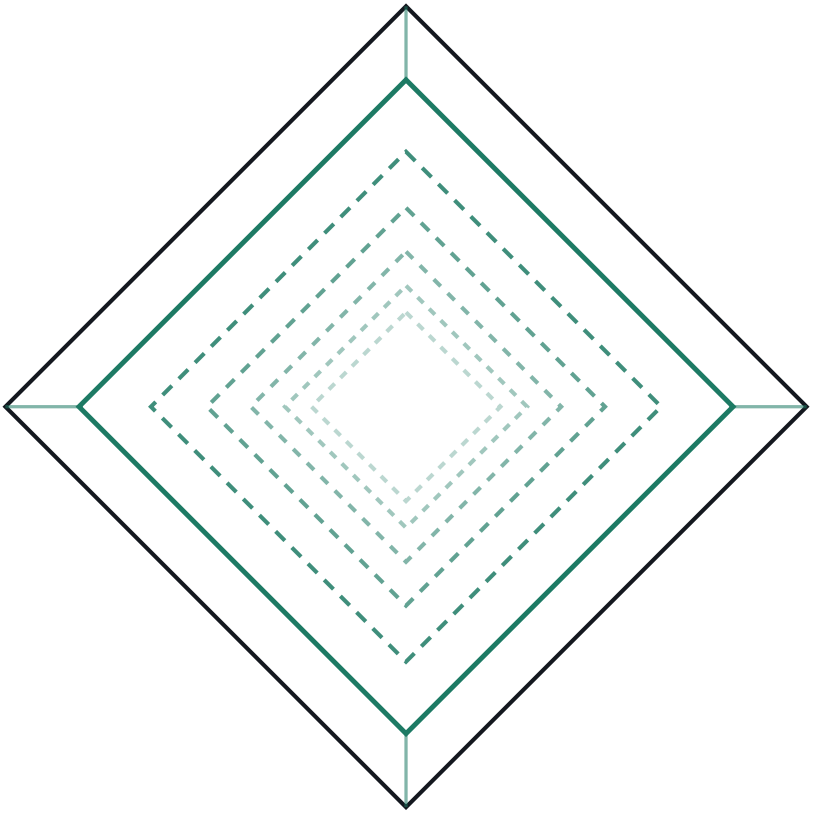
TESSERA

INFINITY LAMPS

DRAFT 0.1 - NOVEMBER 2026



DRAFT FOR FEEDBACK

DIY Kit

KIT BUILD GUIDE · NOT YET AVAILABLE

MADE BY RYAN IN SCOTLAND

About this draft

There is no DIY kit yet. Finished lamps come first. A kit is planned as a follow-up, and before I decide how to make it I would like to hear from the people who might build one.

This document does two jobs:

- it describes what a kit **could** look like, and roughly how the build would go, so you can picture it,
- it asks the questions I need answered before deciding (*Your feedback*, at the end).

Nothing here is a promise of contents, price or date. Parts, steps and tiers will change once I know what builders actually want. Where something is open it says so.

01 What you would end up with

The same lamp as a finished one: a mirrored polyhedron with addressable LEDs along every edge, controlled from any browser, with the same 34 effects, the same sensors, the same optional USB-C power bank and the same free firmware updates. Once it is built, the **User Guide** applies to it exactly as it does to a finished lamp.

What changes is who does which part of the work.

02 How the lamp goes together

Whatever the kit tier, a lamp is built from the same five stages. Each one is a candidate for doing yourself or having done for you.

STAGE	WHAT IT INVOLVES	SKILL AND TOOLS
A. Frame	Printed struts (one per edge) glued into printed corner hubs, held square in a printed glue jig while the glue sets	Patience. Cyanoacrylate glue, the jig, a flat bench
B. LEDs	Cutting LED strip to edge length, laying it into each strut's channel, and soldering the strip ends together at each corner	Confident soldering of small pads. Soldering iron, the printed solder jig
C. Electronics	Fitting the controller board, power socket, power switch, button and power-bank bay into the base, and connecting the LED chain	Screwdriver. The board arrives assembled and tested in every tier
D. Setting up	Telling the lamp its shape and LED layout, then checking every edge with the built-in wiring check	A phone or laptop. No software to install
E. Panels	Cutting acrylic panels, applying mirror film, fitting them into the frame and clipping the covers on	Care and a dust-free room. Acrylic cutting tools, film application kit

The controller board is always supplied **fully assembled, flashed and tested**. It has fine-pitch surface-mount parts, and a bad joint on its power path is invisible and only fails under load. That is not a step I can support at a distance.

03 Possible kit tiers

These are the options I am weighing. The point of the feedback section is to find out which of them anyone wants.

	ASSEMBLY KIT	BUILDER'S KIT	MAKER'S PACK
Controller board, assembled and tested	Included	Included	Included
Power socket, switch, button, bay leads	Included, pre-wired	Included, pre-wired	Included, pre-wired
Frame parts	Printed, frame pre-glued	Printed, loose	Print them yourself from the published files
LED strip	Pre-cut and soldered into a finished chain	Cut to length, you solder the corners	You buy and cut it
Panels	Cut and mirrored	Cut and mirrored	You cut and mirror them, using the printed templates
Glue and solder jigs	Not needed	Printed jigs included	Print them yourself
Firmware	Installed and set up for your lamp	Installed and set up for your lamp	Installed. You enter your own shape and layout
Soldering	None	LED corners only	LED corners only
You need a 3D printer	No	No	Yes

A **power bank is not included in any tier**. It is the only part with lithium cells in it, and you buy it finished and certified (see section 5).

04 Draft build outline

This is the order the build would follow. Detailed steps, photos and part counts will come once the kit contents are settled. Steps marked *(tier)* only apply to kits where you do that stage yourself.

4.1 Check the kit

Unpack onto a soft cloth and check everything against the parts list before you start. Leave the panels in their sleeves until the end: dust or a scratch on a panel is permanent once the lamp is closed.

4.2 Build the frame *(tier)*

1. Dry-fit one strut into one hub to feel how firmly they seat.
2. Glue each strut into its hubs using the glue jig, which holds the joint at the right angle while the glue sets.
3. Build face by face, checking each face sits flat with no gap at any joint.

> ■ WARNING
>
> Glue the frame **before** any panel is unpacked. Cyanoacrylate vapour fogs acrylic permanently, even without touching it.

4.3 Fit the LEDs *(tier)*

1. Lay each length of strip into its strut's channel, LEDs facing into the lamp, following the arrows printed on the strip and the wiring map.
2. Solder the strip ends together at each corner, using the solder jig to hold the two strips at the corner angle.
3. Do not kink the strip. A hard fold cracks its tracks and darkens everything after it.

4.4 Fit the electronics

1. Mount the controller board in the base.
2. Fit the power socket, the power switch and the button, and plug their leads into the board.
3. Connect the start of the LED chain to the board's LED terminal.
4. If your lamp has a power-bank bay, fit the bay lead. Do not fit a bank yet.

4.5 First power-on, with the lamp still open

1. Set the power switch to **OFF**, then plug in a **USB-C PD charger that offers 20 V**. Set the switch to **ON**.
2. Join the lamp's own WiFi network, `infinity-lamp`, and open the control page.
3. Set brightness to about 30%, choose a white palette and *Solid / Gradient*, and look at every edge.

If anything gets warm, smells, or one LED lights very brightly while the rest stay dark, switch off and contact me.

4.6 Set up and check the wiring

1. *(Maker's pack)* On the Advanced settings page, under **LED & shape config**, choose the shape and enter the LEDs per edge and the colour order.
2. Turn on **Wiring check**. It lights one segment at a time: amber, with a green first LED where the data enters and a white dot running the way the data flows. Compare each one with the **Unfold net** view.
3. If a segment runs the wrong way, tick **reversed** for it in the segment mapping. If two are swapped, exchange their edge numbers. Save.

4. Check the sensors: play music and watch the sound meters; tilt the lamp with *Gravity Fill* playing. Then sit the lamp upright and press **Calibrate level**.
5. On the Backup and provisioning panel, press **Use current settings as baseline**, so a future reset returns the lamp to your build.

4.7 Fit the panels *(tier)*

1. Clean hands or cotton gloves. Hold panels by their edges.
2. Lower each panel into its rebate square rather than sliding it in.
3. Clip the covers over the struts around it, working round the face.
4. Look through each panel at a light before moving on. Any dust you can see now will be in the tunnel for good.

4.8 Close up and run it in

1. Fit the base cover. Check no lead is pinched.
2. Run **Auto-cycle** for half an hour. It finds an intermittent connection while the tools are still out.
3. Export a backup from the Advanced settings page and keep the file.

From here the **User Guide** covers everything, including fitting a power bank.

05 The power bank

A kit lamp takes the same optional power bank as a finished one. It is not included. It must have:

- a **USB-C PD output that offers [BANK PD VOLTAGE] V**,
- **20 W** or more,
- a **UKCA or CE mark**, bought from a proper retailer,
- a size that fits the bay: **[BAY DIMENSIONS]**.

One USB-C port is enough, and pass-through charging is not needed. The lamp charges the bank while plugged in and switches to it when the mains goes. User Guide section 13 has the detail.

06 Tools you would need

TOOL	ASSEMBLY KIT	BUILDER'S KIT	MAKER'S PACK
Small screwdriver	Yes	Yes	Yes
USB-C PD charger, 20 V, [CHARGER WATTAGE] W or more	Yes	Yes	Yes
Phone, tablet or laptop with a browser	Yes	Yes	Yes
Soldering iron, fine tip, and solder	-	Yes	Yes

TOOL	ASSEMBLY KIT	BUILDER'S KIT	MAKER'S PACK
Cyanoacrylate glue	-	Yes	Yes
Multimeter	-	Recommended	Recommended
3D printer (PETG, matte black)	-	-	Yes
Acrylic cutting tools and mirror-film kit	-	-	Yes

07 Your feedback

This is the part I need help with. Answer as many as you like, in any order, by email to **[SUPPORT EMAIL]** or as an issue at **github.com/infinity-lamps/tessera/issues** titled *Kit feedback*.

About you

1. Have you built electronics or 3D-printed kits before? Roughly what?
2. Do you own a 3D printer? Which materials do you print?
3. How comfortable are you soldering small pads on LED strip: never tried, could learn, happy doing it?

What the kit should include

4. Which tier in section 3 would you buy, if any? What would you change about it?
5. Would you rather the frame came pre-glued, or build it yourself?
6. Would you rather the panels came cut and mirrored, or cut and film them yourself? Mirror filming is the most skilled and least forgiving step.
7. Should a USB-C power bank be an option in the box, or are you happy to buy your own to the published specification?
8. Is a charger something you would want included?

The lamp itself

9. Which shape, and roughly what size, would you most want to build?
10. Do the microphone and motion sensor matter to you, or would a cheaper kit without them be better?

Instructions and support

11. How would you like the instructions: printed booklet, online guide, PDF, video, or a mix?
12. What would worry you most about building one?
13. What would you expect to pay for the tier you would choose?
14. Anything else you would want me to know?

Thank you. Every answer changes what the kit turns out to be.

Made by Ryan in Scotland

Questions and feedback: **[SUPPORT EMAIL]**

Guides, firmware and print files: **infinity-lamps.github.io/tessera**

Draft for feedback only. No kit is on sale, and the contents described here may change.